

— solved once, offline, from the model —

Passive reference $\hat{y}_{\text{passive}}$

derived from the cooler curve,
not measured — 17.40 vs 17.29 W

DC gain G (27×27)

solved once from $L T = P$
cond 366, $\sigma_1 = 81\%$

$r_{\text{dev}} = r - \hat{y}_{\text{passive}}$

known before the run starts

defines the decoupling

Bounded allocator

$\min \|G u - v\|^2 + \lambda \|u - u_{\text{ref}}\|^2$
 $+ \rho \|u - u_{\text{prev}}\|^2$
subject to $0 \leq u \leq u_{\text{max}}$
project, don't clip

Cryocooler

continuous heat removal

Cryostat

27 heaters, 27 sensors
~24 h dominant mode
8.6 h power $\rightarrow$ temperature lag

27 temps [K]

Measurement filter

$\tau_f = 900$ s

hides the one-step parasitic path, passes the 34 h mode — K_p : 0.09 $\rightarrow$ 5.5

λ : damping, scaled to G 's own spectrum · ρ : rate penalty (0 by default)

The feedforward supplies the holding power; the PI only trims it. Nothing in the live loop inverts G — the allocator solves against it under $u \geq 0$.

